

Apache JMeter

Open-source tool widely used for load and performance testing of protocols and services.

What is it?

JMeter builds test plans with virtual-user groups, requests, timers, assertions and listeners. It is widely used for HTTP/APIs but is not merely a functional API client.

Installation

JMeter depends on Java. Install a compatible JDK, validate java -version, download the official JMeter package and run the appropriate launcher.

Test Plan

Organize Thread Groups, samplers, configuration, data, assertions and listeners. Heavy listeners can distort large tests.

Metrics

Analyze throughput, error rate and percentiles such as p95/p99 together with server CPU, memory, database and dependencies.

Load vs Stress

Load evaluates planned demand; Stress exceeds normal limits to study degradation, capacity and recovery.

Good practices

Define objectives, realistic workload, ramp-up, data, environment, criteria and monitoring before execution.

QA tip

■ QA Tip: performance tests without server observability show symptoms but may hide causes.

Official resources

Site oficial: https://jmeter.apache.org/download_jmeter.cgi